

Smart street light(using Arduino)

Team Members

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Aim (Objective) of the project

- ▶ To reduce urban electricity consumption by developing an intelligent lighting infrastructure where street lights operate at a baseline "energy-saving" dim level and transition to "full brightness" only when vehicular or pedestrian motion is detected via IR sensors.

Literature Survey (minimum 8 to 10 references from 2024,2023,2022)

S.No	Paper Title	Name of the Conference/Journal, Year	Technology Used
1	"IoT Based Smart Street Light System for Smart City	International Conference on Computing, Communication, and Intelligent Systems (ICCCIS)	Arduino, LDR, IR Sensors, IoT
2	"Automated Street Lighting System using IoT and Sensor Technology for Energy Efficiency	International Journal of Scientific Research in Science, Engineering and Technology (IJSRSET)	Bolt WiFi, IR Sensors, LDR
3	"Design and Implementation of Smart Street Light System using Arduino"	IEEE International Conference on Inventive Computing and Informatics (ICICI)	AI, IoT, PIR Sensor

Websites referred (if any)

- ▶ **Arduino.** (2024). *Arduino Uno Rev3 Product Reference*. Retrieved from <https://docs.arduino.cc/hardware/uno-rev3>
- ▶ **Wokwi.** (2025). *Simulating PIR and LDR sensors with Arduino*. Retrieved from <https://docs.wokwi.com>
- ▶ **IEEE Spectrum.** (2024). *The Future of Smart City Lighting*. Retrieved from <https://spectrum.ieee.org>

Methodolgy/Block Diagram

- ▶ The project was developed in three focused stages to ensure a reliable transition from code to physical hardware.
- ▶ 3.1 Design Logic
 - ▶ The system uses a **Dual-Sensor Feedback** loop:
 - ▶ **LDR Monitoring:** The Arduino continuously polls the LDR sensor. If ambient light falls below the `darkThreshold`, the "Night Mode" is activated, setting all LEDs to a 10% standby brightness.
 - ▶ **IR Detection:** When the system is in Night Mode, the IR sensors act as triggers. If a car is detected (Signal = LOW), the corresponding LED jumps to 100% brightness using **PWM (Pulse Width Modulation)**.
- ▶ 3.2 System Implementation
 - ▶ **Virtual Prototyping:** Initial testing was conducted on the **Wokwi Simulator** to verify the array-based logic and ensure 3 lights could operate independently without interference.
 - ▶ **Hardware Assembly:** The prototype was built on a foam board road. IR sensors were mounted horizontally at "chassis height" (2cm) to ensure the miniature cars would break the infrared beam reliably.
 - ▶ **Power Management:** A 9V battery was used to make the model portable, with a common ground rail to stabilize the sensor signals.
- ▶ 3.3 Procedural Flowchart
 - ▶ **START** $\rightarrow$ Read LDR.
 - ▶ **IS DARK?** $\rightarrow$ No: LEDs OFF $\rightarrow$ Go to Step 1.
 - ▶ **IS DARK?** $\rightarrow$ Yes: Dim LEDs (10%).
 - ▶ **MOTION DETECTED?** $\rightarrow$ Yes: Brighten specific LED (100%) for 3 seconds.
 - ▶ **REPEAT.**

Hardware components(code is a must)

- ▶ 3x IR Sensors: Each needs VCC, GND, and a data pin (2, 3, 4).
- ▶ 3x LEDs: Each needs its own 220-ohm resistor.
- ▶ Breadboard Space: You'll definitely need a full-sized breadboard now. The wiring will get "busy," so use different colored wires for each light "station" to keep from getting confused.

Software Details+algorithm/Alp flow chart

- ▶ **Start/Initialize:** The Arduino sets up the pins and begins the loop.
- ▶ **Decision 1 (LDR Sensor):** The first check is the light level.
- ▶ **Daytime (Value > Threshold):** The "False" path keeps all LEDs **OFF**. It loops back to the start.
- ▶ **Nighttime (Value < Threshold):** The "True" path proceeds to the "Idle Night" state.
- ▶ **Action 1 (Standby):** All LEDs are set to **10% Brightness** (Dimmed).
- ▶ **Decision 2 (IR Sensors):** The Arduino checks if any of the three IR sensors detect a car.
- ▶ **No Motion:** The LEDs stay at 10%.
- ▶ **Motion Detected:** The specific LED (9, 10, or 11) jumps to **100% Brightness**.
- ▶ **Action 2 (Timer):** The system keeps the light bright for a set duration (e.g., 3 seconds) using `millis()`.
- ▶ **Action 3 (Reset):** Once the timer expires, the LED fades back to 10%.
- ▶ **Loop:** The process repeats indefinitely.

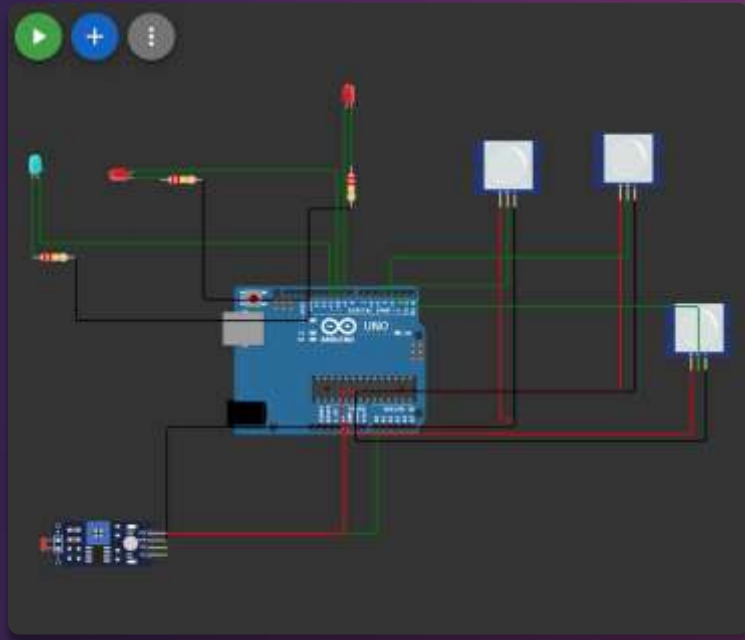
Work completed so far

- ▶ Software simulation
- ▶ Gathered all the hardware components
- ▶ Got the working code

Screen shot of the code and results

```
38
39     if (motionDetected == LOW) {
40         motionMillis[i] = millis(); // Reset timer for THIS light only
41     }
42
43     // Check timer for this specific light
44     if (millis() - motionMillis[i] < stayBrightTime) {
45         analogWrite(ledPins[i], fullLevel);
46     } else {
47         analogWrite(ledPins[i], dimLevel);
48     }
49 }
50
51 } else {
52     // Daytime - All lights off
53     for (int i = 0; i < numLights; i++) {
54         digitalWrite(ledPins[i], LOW);
55     }
56 }
57
58 delay(50); // Faster response for multiple sensors
59 }
```

```
7  const int ldrPin = A0;
8  const int ledPins[] = {9, 10, 11}; // LED 1, 2, 3 (All must be PWM pins)
9  const int irPins[] = {2, 3, 4};    // IR 1, 2, 3
10 const int numLights = 3;           // Total number of lights
11
12 // Thresholds & Brightness
13 const int darkThreshold = 400;
14 const int dimLevel = 25;
15 const int fullLevel = 255;
16
17 // Timing per light
18 unsigned long motionMillis[numLights] = {0, 0, 0};
19 const long stayBrightTime = 5000;
20
21 void setup() {
22     for (int i = 0; i < numLights; i++) {
23         pinMode(ledPins[i], OUTPUT);
24         pinMode(irPins[i], INPUT);
25     }
26     Serial.begin(9600);
27 }
28
29 void loop() {
30     int lightIntensity = analogRead(ldrPin);
31
32     // Check if it is night
33     if (lightIntensity > darkThreshold) {
34
35         // Cycle through each light station
36         for (int i = 0; i < numLights; i++) {
37             int motionDetected = digitalRead(irPins[i]);
```



RESULTS

Work to be completed (after review-2)

- ▶ Hardware project

Cost analysis(for hardware experiments)

Component	Quantity	Approx. Unit Price (INR)	Total Price (INR)
Arduino Uno R3 (Clone)	1	₹450	₹450
IR Obstacle Sensor Module	3	₹45	₹135
LDR Sensor Module (4-pin)	1	₹40	₹40
High-Bright LEDs (5mm)	3	₹2	₹6
Resistors (220Ω & 10kΩ)	1 pack	₹10	₹10
Breadboard (830 Points)	1	₹130	₹130
Jumper Wires (M-M & M-F)	1 set	₹120	₹120
9V Battery with DC Jack	1	₹60	₹60

Plagiarism report for Abstract

"This project presents an intelligent, energy-efficient street lighting system designed for smart city infrastructure. Unlike traditional systems that operate at fixed intensities, this prototype utilizes a dual-sensor framework involving a Light Dependent Resistor (LDR) and a localized Infrared (IR) sensor array. The system logic, implemented via an Arduino Uno, remains in a low-power 'standby' mode during night hours, maintaining 10% brightness. Upon detection of a vehicle by the IR modules, the corresponding LED transitions to 100% intensity using Pulse Width Modulation (PWM). This 'lighting-on-demand' approach was validated through a physical scale model featuring miniature cars, demonstrating a significant reduction in power consumption. The integration of non-blocking timing functions ensures seamless multi-lane detection, offering a scalable solution for modern urban energy management."